YEAR 9 SCIENCE
PHYSICAL SCIENCES
REVISION - LIGHT

1. (a) How do the particles move in a ***transverse wave***? Give an example of this wave.

   (b) How do the particles move in a ***longitudinal wave***? Give an example of this wave.

   (c) Draw a displacement - distance graph to show the following. Make sure you give appropriate scales and labels.

   amplitude = 5.00 cm
   wavelength = 10.0 cm

2. (a) How would you describe ***light*** - what is it?

   (b) Does light need a medium (substance) to travel in? How do you know?

3. (a) List the parts of the electromagnetic spectrum in order.

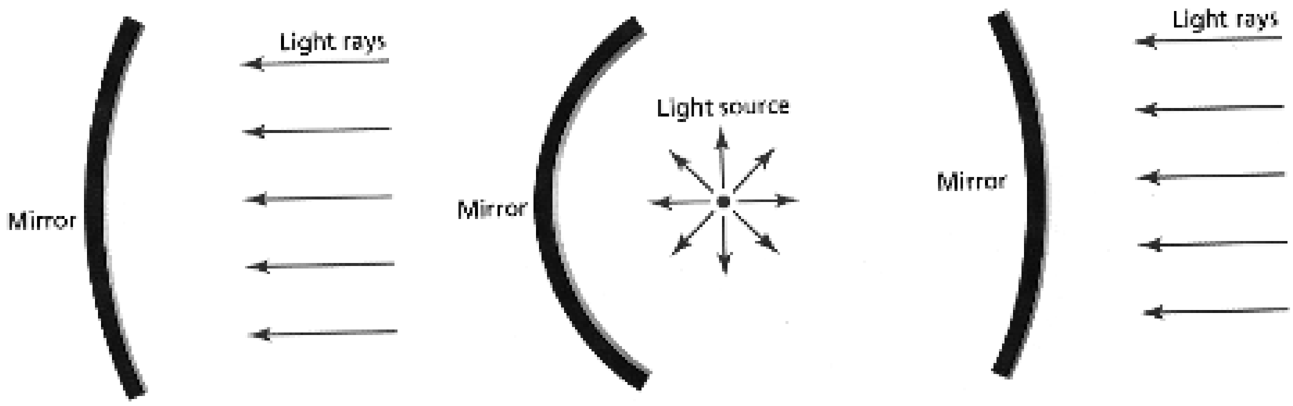
   (b) Give a use or characteristic for each part.

4. (a) What are the Laws of Reflection?

(b) Draw a simple diagram to show it.

5. What is the difference between a ***real image*** and a ***virtual image***?

6. Complete the following ray diagrams.

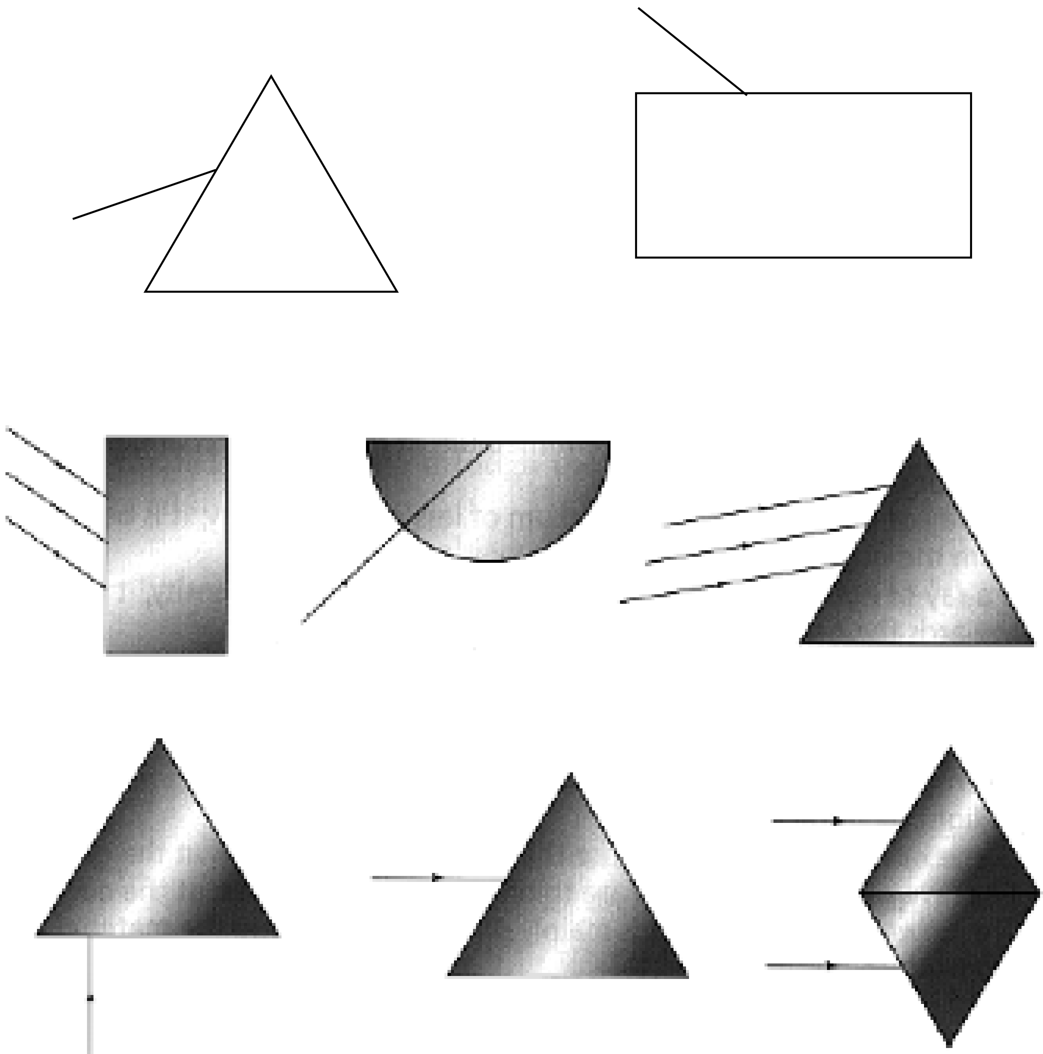


7. (a) Define the term ***refraction***.

(b) Will a light ray refract when it enters a new medium ***at right angles*** (along the normal)?

(c) If a light ray enters a new medium where it *speeds up*, does it refract *towards or away from* the normal?

8. Complete the following ray diagrams to show the path taken through the glass prisms.



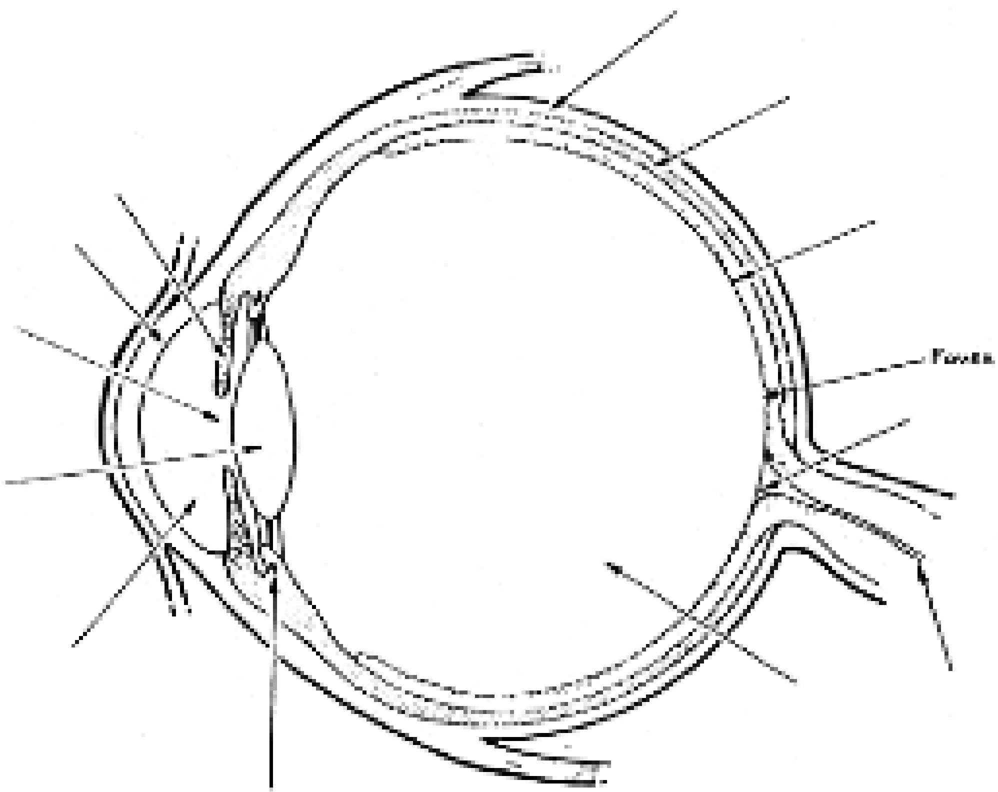
9. (a) If a person on a jetty wishes to spear a fish swimming in the water (not directly below), where should they aim?

(b) Draw a simple diagram to show this situation.

10. (a) If an object is placed a long way from a convex lens, what type of image is formed?

(b) Where should an object be placed when a convex lens is used as a magnifying glass?

11. (a) Label the parts of the eye on the diagram below.



(b) Give the major function for each part.

12. Describe how the lens of the eye changes to focus on a ***close object*** and a ***distant object***.

13. What is the ***blind spot*** in the eye?